

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3

Duration : 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

Code of Conduct:

I R. Akshaya (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



ARTIFICIAL INTELLIGENCE

COI - AI

ANALYTICAL PROBLEM SOLVING.

Student 3: Akshaya R

Register No: 192584118

① An Exploration robot is sent into an unknown cave system with no prior map, and its sensors have limited range. Explain how the robot can use search with partial information to build a map while searching for an exit, and discuss the risk of irreversible actions in the environment.

Sol:- * An Exploration robot is an unknown cave has no prior map and can only observe nearby areas using limited-range sensors.
* In this situation, it uses online (contingency) search, where it senses the environment, decides the next move, executes the action, and update its knowledge continuously.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Excellent)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

95
100

Student 3: Akshaya R | Register Number: 192524118

1. An exploration robot is sent into an unknown cave system with no prior map, and its sensors have limited range. Explain how the robot can use search with partial information (contingency/online search) to build a map while searching for an exit, and discuss the risks of irreversible actions in this environment.
2. Solve the 6-Queens problem: place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the problem components (state space, initial state, successor function, goal test) and explain how incremental formulation reduces the search space compared to complete-state formulation.
3. A hospital wants to build an AI agent to triage incoming patients based on symptom severity. Define an appropriate performance measure for this agent and explain the concept of rationality in this context — under what conditions would the agent's action be considered rational?

Rationality :

A rational agent selects action that maximizes the Expected performance based on the available information.

The agent is considered rational when it:

- ⇒ Uses patient symptoms and medical history.
- ⇒ Correctly estimate the severity level.
- ⇒ Prioritize critical patients immediately.
- ⇒ Follows medical guidelines.
- ⇒ Updates decisions when new information

Example: If a patient has severe chest pain, low blood pressure, and difficulty breathing, a rational AI agent immediately assigns high priority for emergency treatment instead of making the patient wait.

Conclusion: A rational hospital triage agent makes decision that maximize patient safety, improve treatment quality and efficiently utilize hospital resources.

3. A hospital wants to build an AI agent to triage incoming patients based on symptom severity. Define an appropriate performance measured for this agent. and Explain the concept of rationality in this context - under what conditions would the agent's action be considered rational?

Ans :- AI Agent for Hospital Patient Triage:

Performance Measure:

The AI agent should:

- * Correctly identify emergency patients.
- * Minimize waiting time.
- * Reduces incorrect diagnosis.
- * Prioritize patients based on severity.
- * Improve patient safety and treatment efficiently.

Goal Test:

8 queens are placed safely on the board without any conflicts.

Incremental Formulation:

In incremental formulation:

1. Queens are placed one row at a time.
2. Unsafe positions are rejected immediately.
3. Only valid partial solutions are expanded.

Advantages Over Complete-State Formulation:

- * Reduces search space significantly.
- * Avoids checking all $6^6 = 46,656$ possible board arrangements.
- * ~~Faster~~ and more memory-efficient.

Row	Column
4	1
5	3
6	5

No two queens attack each other horizontally, vertically, or diagonally.

Problem Formulation:

State Space:

All possible arrangements of queens on a 6×6 chessboard.

Initial State:

Empty chessboard (no queens placed).

Success Function:

Place one queen in the next row in a safe column where it is not attacked by any previously placed queen.

Conclusion :

⇒ Online Search Enables that robot to explore unknown environments while learning from Experience.

⇒ However, irreversible actions may trap the robot or prevent it from reaching the Exit.

2. Solve the 6-Queens problem: place 6 queens on a 6x6 chessboard such that no two queens attack each other. Formulate the problem components (state space, initial state, successor function, goal test) and explain how incremental formulation reduces the space search compared to complete state formulation.

Ans:-

One Valid Solution:

Row	Column
1	2
2	4
3	6

Working :-

- * The robot starts from the Entrance.
- * It scans nearby paths using its sensors.
- * It record to an unexplored location.
- * It records visited path and builds an internal map.
- * If it reaches a dead End, it backtracks and explores another route.
- * The process continues until the Exist is found.

Map Building:-

- * Stores visited locations
- * Marks blocked paths and dead Ends.
- * Updates connections between Explored areas
- * Gradually creates a complete map of cave

Risks of irreversible Actions:-

- * Entering a one way tunnel.
- * Falling into a pits or unstable areas
- * Closing passages permanently.
- * Running out of battery before finding the Exist.